Nabeel Sherazi | Senior Autonomy Engineer | New York City, NY 11101 | 203-306-6399
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Senior autonomy engineer with 6+ years of experience shipping real robots to customers. Deep domain knowledge from shipping cellular teleoperated passenger cars, stratospheric autonomous gliders, vertical takeoff jets. Machine learning, multi-agent motion planning, perception, hard realtime controls, HIL, embedded Linux, and embedded microcontrollers (bare metal and RTOS).
Languages I prefer to work in: C++, C, Python
Languages I have working ability with: Go, Rust, CUDA, Typescript, C#, MATLAB

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Mach Industries (2024 - 2025)

- Enabled GNSS-denied visual-inertial navigation on our high altitude autonomous glider via ground-up sensor fusion stack on Jetson Orin NX: camera selection, custom drivers, sub-1ms time synchronization infrastructure for partner VIO/feature-matching pose estimator integration.
- Developed safety-critical realtime flight logic in custom PX4 module (C++) for Viper (Group 3 VTOL cruise jet) under a two week time constraint: mission planning, landing guidance, and release sequencing; validation prior to first flight via extensive SITL and HITL testing.
- Represented engineering at Army Applications Lab test range events. Live debugging, technical leadership engagement, and contributing to a successful demonstration that closed a multi-million dollar production contract.
- Led EMI characterization and mitigation for USB3 camera array on Glide: coordinated VNA measurements with RF team, diagnosed chassis noise coupling, and redesigned sensor integration to meet RF compatibility requirements.
- Led development of perception-aware digital twin simulation environment in Unreal Engine + Cesium for physically, geographically, and optically accurate automated autonomy testing; directed implementation via daily technical guidance, then assumed full ownership and maintenance post-project.
- Bootstrapped Jetson Linux OS build pipeline in CI using QEMU, enabling reproducible cross-platform builds with custom package integration across the fleet.
- Implemented parallelized automated airgapped fleet deployment with Ansible and Docker, reducing deploy time to 10 minutes for a fleet of 10 Glides, enabling rapid iteration.
- Hiring manager and interviewer for software, embedded, and embedded Linux roles. Managed a team of three direct report interns and one full-time direct report.

Software Engineer, Halo.car (2022 - 2024)

- Greenfield-designed Komodo telematics system on STM32H4: CAN bus vehicle telemetry ingestion, LTE-M connectivity, and MQTT publishing for real-time fleet health and location monitoring.
- Built GCP backend for Komodo fleet telemetry: MQTT broker, Python message aggregation service, and database pipeline feeding real-time operations dashboards across 15 vehicle fleet for 6 months of operation.
- Designed and wrote the write-ahead logging system (C++) for our safety-critical watchdog, Cerberus, enabling us to pinpoint subsystem failures even under adverse network conditions or loss of power.
- Implemented a custom UVC Linux driver to obtain 50% more bandwidth efficient compressed video streaming from cameras, allowing higher resolution video and eliminating bandwidth-saturation USB controller critical failures.
- Refactored GStreamer teleoperation pipeline to reduce thread contention, reducing mean processing latency across ODD by 10ms.
- Deployed realtime CV pipeline (YOLO) for traffic light detection in teleop HUD; architected ROS/GStreamer integration with extensibility for incremental L2 autonomy.

Software Engineer, Autonodyne (2019 - 2022)

- Extended RT-RRT (dynamic single-agent 2D path planner) to centralized multi-agent 3D replanning, novel adaptation handling real-time obstacle avoidance across N agents with shared tree structure.
- Built Aviary, a multi-agent UAS simulation harness integrating AirSim/Gazebo physics with ArduPilot/PX4 autopilot instances, enabling repeatable validation of multi-agent planning algorithms.
- Developed a Unity-based demo of multi-agent RRT*, demonstrating scalability with 25 agents running concurrently on mobile hardware.

EDUCATION

Northeastern University, B.S. Applied Physics (2020)

SKILLS

- Embedded software: UART , SPI, I2C, CAN, DDS, RS-232, NVIDIA Jetson, STM32 family, FreeRTOS, NuttX
- Motion Planning & Autonomy: ROS2, sampling based methods, trajectory optimization, mission planning, SLAM
- Perception & Sensor Fusion: Cameras, GNSS, IMU, Barometers, Airspeed
- Simulation: NVIDIA Isaac Sim / Omniverse, AirSim, CARLA, Unity and Unreal Engine